

The Subnet Mask

Function of the subnet mask

The subnet mask determines how many bits are reserved for the network portion (represented by 1s) and the host portion (represented by 0s). To separate the network and the host portions, a bitwise AND operation is used.

Binary Conversion

Let's examine the network **192.168.1.0/24** with the subnet mask **255.255.255.0** and work with the IP **192.168.1.5**. Since subnet masks operate in binary, we first need to convert everything into binary.

192 . 168 . 1 . 5 → 11000000 . 10101000 . 00000001 . 00000101

255 . 255 . 255 . 0 → 11111111 . 11111111 . 11111111 . 00000000

These two binary numbers are then "added" together via a bitwise AND operation.

What is a Bitwise AND Operation?

A bitwise AND operation is a logical check where both inputs must be true (1) to return 1.

1	AND	1	=	1
1	AND	0	=	0
0	AND	1	=	0
0	AND	0	=	0

Applying the Bitwise AND Operation

The result of the bitwise AND operation is called the network ID, which is always the first address in a network and is unusable, because it defines the network itself.

```
1 1 0 0 0 0 0 0 . 1 0 1 0 1 0 0 0 . 0 0 0 0 0 0 0 1 . 0 0 0 0 0 1 0 1
1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 0 0 0 0 0 0 0 0
------(AND)-
1 1 0 0 0 0 0 0 . 1 0 1 0 1 0 0 0 . 0 0 0 0 0 0 0 1 . 0 0 0 0 0 0 0 0
                        (Network portion) | (Host portion)
                                      split here
```

The split occurs at this position because the subnet mask is **255.255.255.0**.

Decimal Conversion

The network ID **11000000.10101000.00000001.00000000** converts to **192.168.1.0**, matching the original network definition (**192.168.1.0/24**).

Examples

172.16.1.42 AND 255.255.255.0 = 172.16.1.0

172.16.1.35 AND 255.255.255.0 = 172.16.1.0

These addresses are in the same network and can communicate directly.

172.16.5.23 AND 255.255.255.0 = 172.16.5.0

172.16.8.23 AND 255.255.255.0 = 172.16.8.0

These addresses are in different networks and require a router to communicate.

Larger Networks

Let's take the network **192.168.0.0/22**. The subnet mask of this network is: **255.255.252.0**. This is a network with a total of 1024 IP addresses from which are 1022 usable. The network range is: **192.168.0.0 - 192.168.3.255**.

We take an IP from the center: **192.168.2.5**

```
1 1 0 0 0 0 0 0 . 1 0 1 0 1 0 0 0 . 0 0 0 0 0 0 1 0 . 0 0 0 0 0 1 0 1
1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 0 0 . 0 0 0 0 0 0 0 0
-----
1 1 0 0 0 0 0 0 . 1 0 1 0 1 0 0 0 . 0 0 0 0 0 0 0 0 . 0 0 0 0 0 0 0 0
```

So the Network ID is **11000000.10101000.00000000.00000000** or in decimal **192.168.0.0**

The same operation with the IP **192.168.5.5**, which should be in another network.

```
1 1 0 0 0 0 0 0 . 1 0 1 0 1 0 0 0 . 0 0 0 0 0 1 0 1 . 0 0 0 0 0 1 0 1
1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 0 0 . 0 0 0 0 0 0 0 0
-----
1 1 0 0 0 0 0 0 . 1 0 1 0 1 0 0 0 . 0 0 0 0 0 1 0 0 . 0 0 0 0 0 0 0 0
```

This network ID is **11000000.10101000.00000100.00000000** which results in **192.168.4.0** in decimal. Since **192.168.3.255** is the last IP address of the **192.168.0.0/22** network, and thus the broadcast address, **192.168.4.0** is the next higher address and thus defines the next network. e.g. **192.168.4.0/24**

!!!CAUTION!!! The first address of a network is always the network ID, and the last address is always the broadcast address. These are unusable for us! Only addresses from the second to the penultimate are usable (e.g., **192.168.0.1 - 192.168.3.254** in **192.168.0.0/22**).